



Inspiring Excellence

Assignment-04

Makeup Assignment-01

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Section: 13

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Ans. to the ques. no-01

a

$$P_2(x) = a_0 + a_1x + a_2x^2$$

$$P_2(0) = a_0 ; a_0 = 3$$

$$P_2(-1) = a_0 - a_1 + a_2 ; a_0 - a_1 + a_2 = 7$$

$$P_2(1) = a_0 + a_1 + a_2 ; a_0 + a_1 + a_2 = -2$$

$$P_2(3) = a_0 + 3a_1 + 9a_2 ; a_0 + 3a_1 + 9a_2 = -4$$

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & -1 & 1 \\ 1 & 1 & 1 \\ 1 & 3 & 9 \end{bmatrix} ; b = \begin{bmatrix} 3 \\ 7 \\ -2 \\ -4 \end{bmatrix}$$

$$u_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} ; u_2 = \begin{bmatrix} 0 \\ -1 \\ 1 \\ 3 \end{bmatrix} ; u_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 9 \end{bmatrix}$$

$$P_k = U_k - \sum_{j=1}^{k-1} (u_k^T q_j) q_j$$

Step-1: ($k=1$)

$$P_1 = U_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$q_1 = \frac{1}{\sqrt{1^2+1^2+1^2+1^2}} \times \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$$

Step-2: ($k=2$)

$$P_2 = \begin{bmatrix} 0 \\ -1 \\ 1 \\ 3 \end{bmatrix} - \left([0 \ -1 \ 1 \ 3] \times \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} \right) \times \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ -1 \\ 1 \\ 3 \end{bmatrix} - \left(\frac{3}{2} \times \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} \right)$$

$$= \begin{bmatrix} 0 \\ -1 \\ 1 \\ 3 \end{bmatrix} - \begin{bmatrix} \frac{3}{4} \\ \frac{3}{4} \\ \frac{3}{4} \\ \frac{3}{4} \end{bmatrix}$$

$$= \begin{bmatrix} -\frac{3}{4} \\ -\frac{7}{4} \\ \frac{1}{4} \\ \frac{9}{4} \end{bmatrix}$$

$$q_2 = \frac{1}{\sqrt{\frac{9}{16} + \frac{49}{16} + \frac{1}{16} + \frac{81}{16}}} \begin{bmatrix} -\frac{3}{4} \\ -\frac{7}{4} \\ \frac{1}{4} \\ \frac{9}{4} \end{bmatrix}$$

$$= \begin{bmatrix} -\frac{3\sqrt{35}}{70} \\ -\frac{\sqrt{35}}{10} \\ \frac{\sqrt{35}}{70} \\ \frac{9\sqrt{35}}{70} \end{bmatrix}$$

Step-3

$$K=3$$

$$P_3 = U_3 - \left\{ (U_3^T q_1) q_1 + (U_3^T q_2) q_2 \right\}$$

$$= \begin{bmatrix} 0 \\ 1 \\ 1 \\ 9 \end{bmatrix} - \left\{ \left(\begin{bmatrix} 0 & 1 & 1 & 9 \end{bmatrix} \times \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} \right) \times \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} + \right.$$

$$\left. \left(\begin{bmatrix} 0 & 1 & 1 & 9 \end{bmatrix} \times \begin{bmatrix} -\frac{3\sqrt{35}}{70} \\ -\frac{\sqrt{35}}{70} \\ \frac{\sqrt{35}}{70} \\ \frac{9\sqrt{35}}{70} \end{bmatrix} \right) \times \begin{bmatrix} -\frac{3\sqrt{35}}{70} \\ -\frac{\sqrt{35}}{10} \\ \frac{\sqrt{35}}{70} \\ \frac{9\sqrt{35}}{70} \end{bmatrix} \right\}$$

$$= \begin{bmatrix} 0 \\ 1 \\ 1 \\ 9 \end{bmatrix} - \left\{ \frac{11}{2} \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} + \frac{15\sqrt{35}}{14} \begin{bmatrix} -\frac{3\sqrt{35}}{70} \\ -\frac{\sqrt{35}}{10} \\ \frac{\sqrt{35}}{70} \\ \frac{9\sqrt{35}}{70} \end{bmatrix} \right\}$$

$$= \begin{bmatrix} 0 \\ 1 \\ 1 \\ 9 \end{bmatrix} - \begin{bmatrix} \frac{11}{4} \\ \frac{11}{4} \\ \frac{11}{4} \\ \frac{11}{4} \end{bmatrix} - \begin{bmatrix} -\frac{45}{28} \\ -\frac{15}{4} \\ \frac{15}{28} \\ \frac{135}{28} \end{bmatrix}$$

$$= \begin{bmatrix} -\frac{8}{7} \\ 2 \\ -\frac{16}{7} \\ \frac{10}{7} \end{bmatrix}$$

$$; q_3 = \frac{1}{\frac{2\sqrt{154}}{7}}$$

$$\begin{bmatrix} -\frac{8}{7} \\ 2 \\ -\frac{16}{7} \\ \frac{10}{7} \end{bmatrix}$$

$$= \begin{bmatrix} -\frac{2\sqrt{154}}{77} \\ \frac{\sqrt{154}}{22} \\ -\frac{4\sqrt{154}}{77} \\ \frac{5}{\sqrt{154}} \end{bmatrix}$$

$$Q = \begin{bmatrix} \frac{1}{2} & \frac{-3\sqrt{35}}{70} & \frac{-2\sqrt{154}}{77} \\ \frac{1}{2} & \frac{-\sqrt{35}}{10} & \frac{\sqrt{154}}{22} \\ \frac{1}{2} & \frac{\sqrt{35}}{70} & \frac{-4\sqrt{154}}{77} \\ \frac{1}{2} & \frac{9\sqrt{35}}{70} & \frac{5}{\sqrt{154}} \end{bmatrix}$$

C

$$R = \begin{bmatrix} U_1^T q_1 & U_2^T q_1 & U_3^T q_1 \\ 0 & U_2^T q_2 & U_3^T q_2 \\ 0 & 0 & U_3^T q_3 \end{bmatrix}$$

$$U_1^T q_1 = \begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix} \times \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} = 2$$

$$U_2^T q_1 = \begin{bmatrix} 0 & -1 & 1 & 3 \end{bmatrix} \times \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} = \frac{3}{2}$$

$$U_3^T q_1 = \begin{bmatrix} 0 & 1 & 1 & 9 \end{bmatrix} \times \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} = \frac{11}{2}$$

$$U_2^T q_2 = \begin{bmatrix} 0 & -1 & 1 & 3 \end{bmatrix} \times \begin{bmatrix} \frac{-3\sqrt{35}}{70} \\ \frac{-\sqrt{35}}{10} \\ \frac{\sqrt{35}}{70} \\ \frac{9\sqrt{35}}{70} \end{bmatrix} = \frac{\sqrt{35}}{2}$$

$$U_3^T q_2 = \begin{bmatrix} 0 & 1 & 1 & 9 \end{bmatrix} \times \begin{bmatrix} -\frac{3\sqrt{35}}{70} \\ -\frac{\sqrt{35}}{10} \\ \frac{\sqrt{35}}{70} \\ \frac{9\sqrt{35}}{70} \end{bmatrix}$$

$$= \frac{15\sqrt{35}}{14}$$

$$U_3^T q_3 = \begin{bmatrix} 0 & 1 & 1 & 9 \end{bmatrix} \times \begin{bmatrix} -\frac{2\sqrt{159}}{77} \\ \frac{\sqrt{159}}{22} \\ -\frac{4\sqrt{159}}{77} \\ \frac{5}{\sqrt{159}} \end{bmatrix}$$

$$= 3.55$$

$$\therefore R = \begin{bmatrix} 2 & \frac{3}{2} & \frac{11}{2} \\ 0 & \frac{\sqrt{35}}{2} & \frac{15\sqrt{35}}{14} \\ 0 & 0 & 3.55 \end{bmatrix}$$

$$R_n = \begin{bmatrix} 2 & \frac{3}{2} & \frac{11}{2} \\ 0 & \frac{\sqrt{35}}{2} & \frac{15\sqrt{35}}{14} \\ 0 & 0 & 3.55 \end{bmatrix} \times \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix}$$

$$= \begin{bmatrix} 2a_0 + \frac{3}{2}a_1 + \frac{11}{2}a_2 \\ \frac{\sqrt{35}}{2}a_1 + \frac{15\sqrt{35}}{14}a_2 \\ 3.55a_2 \end{bmatrix}$$

$$QT \cdot b = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ -\frac{3\sqrt{35}}{70} & -\frac{\sqrt{35}}{10} & \frac{\sqrt{35}}{70} & \frac{9\sqrt{35}}{70} \\ -\frac{2\sqrt{154}}{77} & \frac{\sqrt{154}}{22} & \frac{-4\sqrt{154}}{77} & \frac{5}{\sqrt{154}} \end{bmatrix} \times \begin{bmatrix} 3 \\ 7 \\ -2 \\ -4 \end{bmatrix}$$

$$= \begin{bmatrix} 2 \\ -8.113 \\ 2.6592 \end{bmatrix}$$

e

$$R_n = Q^T b$$

$$\begin{bmatrix} 2a_0 + \frac{3}{2}a_1 + \frac{11}{2}a_2 \\ \frac{\sqrt{35}}{2}a_1 + \frac{15\sqrt{35}}{14}a_2 \\ 3.55a_2 \end{bmatrix} = \begin{bmatrix} 2 \\ -8.113 \\ 2.6592 \end{bmatrix}$$

$$\therefore 3.55a_2 = 2.6592$$

$$\therefore a_2 = 0.749$$

$$\frac{\sqrt{35}}{2}a_1 + \frac{15\sqrt{35}}{14}a_2 = -8.113$$

$$\therefore a_1 = -4.348$$

$$2a_0 + \frac{3}{2}a_1 + \frac{11}{2}a_2 = 2$$

$$\therefore a_0 = 2.20$$

$$\therefore a_0 = 2.20; a_1 = -4.348; a_2 = 0.749$$

$$\therefore P_2(x) = 2.20 - 4.348x + 0.749x^2$$